How-To Library

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How to Configure Hive ODBC Connector for Cloudera Enterprise with PowerCenter on Linux

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Abstract

You can configure Hive ODBC Connector for Cloudera Enterprise with PowerCenter® on Linux. This article describes how to install and configure the connector and also use the connector with the PowerCenter installation deployed in your infrastructure.

Supported Versions

- PowerCenter 10.x , 10.1.x, and 10.2

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Hive ODBC Connector for Cloudera Enterprise Overview

With Hive ODBC Connector for Cloudera Enterprise, the PowerCenter Integration Service can read data from Hive data sources or write data to multiple Hive data targets.

Hive ODBC Connector for Cloudera Enterprise is available for Microsoft Windows, Linux, IBM AIX, and Mac OS X. The driver complies with the ODBC 3.80 data standard and includes Unicode and 32-bit and 64-bit support for high-performance computing environments on all platforms.

Install Hive ODBC Connector for Cloudera Enterprise Driver on Linux

You can install Hive ODBC Connector for Cloudera Enterprise Driver on Linux to connect to Hive using PowerCenter. With the help of the Cloudera Hive ODBC driver, you can read from Hive sources and write to Hive targets in PowerCenter.

1. Download and install Hive ODBC Connector for Cloudera Enterprise driver for Linux from the following Cloudera link:

   https://www.cloudera.com/downloads/connectors/hive/odbc/2-5-24.html
2. Add the Hive odbc bin directory to the PATH environment variable.

   Using a Bourne shell:

   ```
   $ PATH=${PATH}:${Hive_ODBC_installed_location}/bin; export PATH
   ```

   Using a C shell:

   ```
   $ setenv PATH ${PATH}:{Hive_ODBC_installed_location}/bin
   ```
3. Add lib to the library path variable.

   Using a Bourne shell:

   ```
   $LD_LIBRARY_PATH=${LD_LIBRARY_PATH}:${Hive_ODBC_installed_location}/lib; export
   LD_LIBRARY_PATH
   ```

   Using a C shell:

   ```
   $ setenv LD_LIBRARY_PATH ${LD_LIBRARY_PATH}:{Hive_ODBC_installed_location}/lib
   ```

4. Optionally, copy the ODBC.ini file to the home directory or another location.

   Default is the Informatica home directory.

   ```
   cp $ODBCHOME/ODBC.ini $HOME/ODBC.ini
   ```

5. Set the ODBCINI environment variable to the location of the ODBC.ini file with the following syntax:

   ```
   $ ODBCINI=$ODBCHOME/ODBC.ini; export ODBCINI
   ```

6. Edit the ODBC.ini file specified by the $ODBCINI environment variable.

   a. Set the InstallDir to the installed location of the Hive ODBC driver.

      For example, `InstallDir= /opt/cloudera/hiveodbc`

   b. Add an entry for the ODBC data source (DSN) under the section [ODBC Data Sources] and configure the data source.

      The following sample is of a DSN entry configured for a Hadoop cluster that uses Kerberos authentication, but without SSL authentication:

      ```
      Cloudera Hive ODBC Driver 64-bit]
      Description=Cloudera ODBC Driver for Apache Hive (64-bit)
      Driver=/opt/cloudera/hiveodbc/lib/64/libclouderahiveodbc64.so
      HiveServerType=2
      Host=192.168.222.160
      Port=10000
      AuthMech=1
      ThriftTransport=1
      KrbHostFQDN=krbhost.krb.com
      KrbServiceName=hive
      KrbRealm=KRB.KDEV.COM
      ```

      For more information about multiple ODBC configuration parameters, see the Cloudera documentation in the following link:

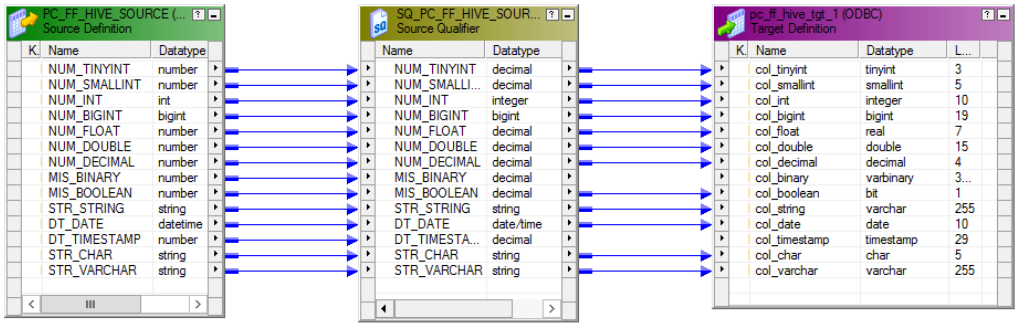
      http://www.cloudera.com/documentation/other/connectors/hive-odbc/latest/Cloudera-ODBC-Driver-for-Apache-Hive-Install-Guide.pdf

   c. Restart the Informatica services.

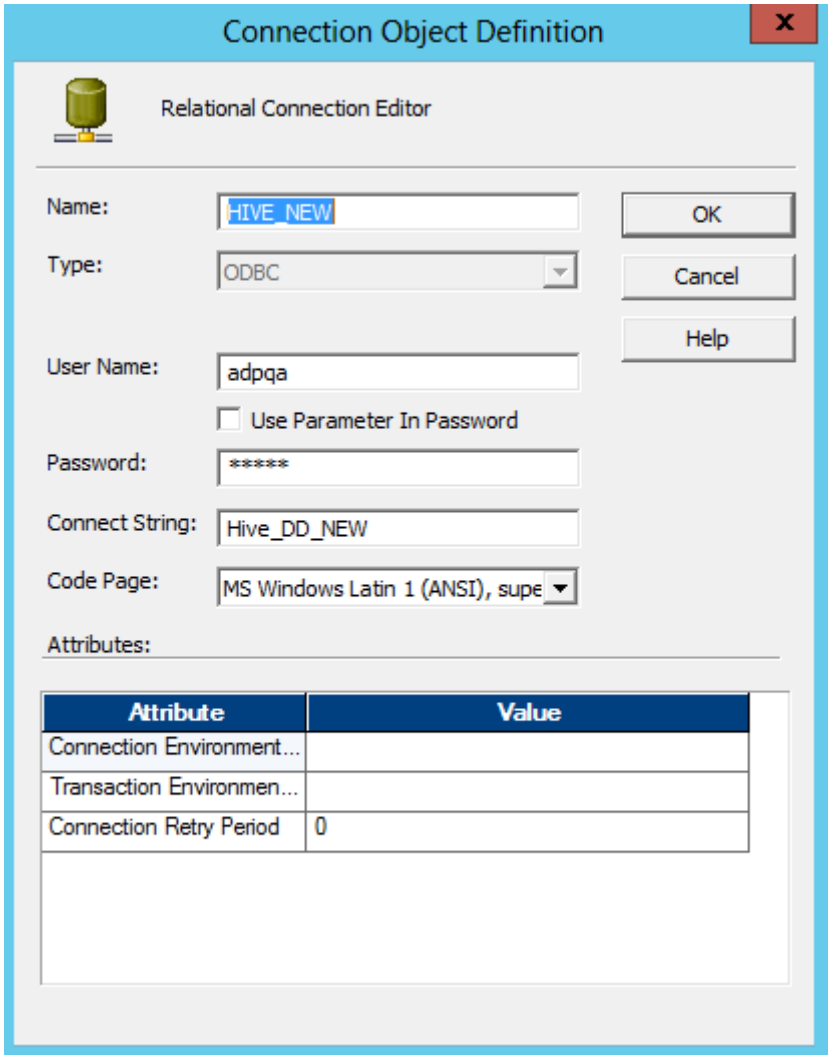
Tips on PowerCenter Configuration

After you install the Hive ODBC Connector for Cloudera Enterprise, you can create a PowerCenter mapping that connects to a Hive source or target using a Hive ODBC connection. For example, in the PowerCenter Designer, create a mapping that reads from an Oracle source and writes to a Hive target.

The following diagram shows a sample mapping:



In the Workflow Manager under Connection Object Definition, create an ODBC connection. Configure the ODBC connection to point to the Hive DSN.



Note: PowerCenter supports only ODBC compliant functionality of the Hive ODBC driver.

For more information about connecting to an ODBC data source, see the *Informatica Installation and Configuration Guide* at the following link:

https://docs.informatica.com/data-integration/powercenter/10-2/installation-and-configuration-guide/installation-overview.html.

For more information about the Hive ODBC connector for Cloudera Enterprise, refer the Cloudera documentation in the following link:

https://www.cloudera.com/downloads/connectors/hive/odbc.html

Troubleshooting

PowerCenter fails to write to a Hive partitioned table and bucketed tables fail.

When you create a partitioned table, Hive expects the following syntax format to insert the data into table:

```
INSERT INTO TABLE pageviews PARTITION (datestamp = '2014-09-23')
VALUES ('jsmith', 'mail.com', 'sports.com'), ('jdoe', 'mail.com', null);
```

Insert query expects partition and partitioned columns to be included in the query. PowerCenter generates the ANSI standard DML query. No workaround exists for the issue.

PowerCenter cannot use the truncate target table property in the session properties.

ODBC generates a delete query when you have selected the truncate table property. All databases support the Delete statement, which is an ANSI compliant SQL. Only few databases supports truncate table, which are not an ANSI compliant option, such as Oracle, Netezza, and Hive. ODBC adapter is generic and is built to support ANSI compliant SQL.

To resolve this issue, you can use the pre sql option in the session to include a `truncate table` *`tname`* statement. The pre sql truncates the table before the data loading begins.

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